RESIDUAL GAP CLASSES IN THE EVEN-DEGREE FERMAT-FOURFOLD CENSUS THROUGH DEGREE 250: EXCHANGE WALLS AND TWO CLOSURES

RIFAT JUMAGULOV

ABSTRACT. For the Fermat fourfold $X_m^4 : x_0^m + \cdots + x_5^m = 0$, the companion note [6] proves that the sector beyond the classical algebraicity machinery is empty in odd degree ≤ 199. This paper is the systematic treatment of the *even*-degree sector through degree 250, which behaves differently: gap classes are frequent, and the odd law "survivor $\Rightarrow 3 \mid m$" fails. We first identify the working lattice with Aoki's group S_m generator-by-generator — including his $p = 2$ standard elements, which we show are identically grade-2 Hodge quadruples, hence Lefschetz-algebraic — and prove a negation-closure lemma converting any integer membership certificate into a literal Aoki claim chain. The even-parity census — run under the note's completeness lemma, which applies to both parities; receipts and independent verifiers in the ancillary — finds forty primitive classes that survive the four base predicates and the complete depth-≤ 2 closure oracle through degree 250; deeper partition certificates close all but ten. An exchange theorem generates claim-equivalences, and an exhaustive scan organizes the ten into five walls. Two of the five then close, both by a coset-transfer lemma: the $m = 168$ class is congruent modulo S_{168} to an explicit $*$-split class, and the isolated degree-210 class lies in S_{210} itself, by an explicit 40-generator ± 1 certificate displayed in full. The residual even-degree boundary at $m \leq 250$ — the classes unresolved by the implemented closure family and the cited results of the two-paper package — is eight certified gap classes ($\nu \notin S_m$, $2\nu \in S_m$) in three exchange walls W_{70}, W_{110}, W_{114}; every lattice verdict carries an independent certificate (modular kernel witnesses for non-membership, exact re-summation for membership).

1. Introduction

Let $X_m^4 \subset \mathbb{P}^5$ be the Fermat fourfold, $G = \mu_m^6/\mu_m$, and for a character $a = (a_0, \ldots, a_5) \in (\mathbb{Z}/m)^6$ with $\sum a_i \equiv 0$, $a_i \neq 0$, let $V(a) \subset H^4_{\text{prim}}$ be the one-dimensional eigenspace; a is a Hodge $(2,2)$ character iff $\sum_i \langle ta_i \rangle = 3m$ for all $t \in (\mathbb{Z}/m)^\times$ (Shioda [8]). The classical algebraicity machinery — Shioda's inductive structure with da Silva's quasi-decomposability formalism [8, 11, 5] and Aoki's standard cycles [1] — and the notation of this paper are those of the companion note [6], whose conventions we use throughout: characters are *ordered* tuples; claim(a) ($V(a)$ spanned by algebraic cycle classes) is invariant under coordinate permutations (automorphisms of X_m^4) and is a Galois-orbit-level property; the census classifies sorted representatives modulo both actions; *grade* g means $\sum_i \langle ta_i \rangle = gm$ for every unit t; $*$ is juxtaposition of entry tuples, $\uplus$ multiset union; χ_a is the Jacobi-sum Hecke character of a and $u(a,p)$ its Tate-normalized value at a split prime p [12, 14], while $\nu(a) \in \mathbb{Z}^{m-1}$ is Aoki's multiplicity vector, additive under juxtaposition (he writes u for ν); and the inflation/descent lemmas of [6] make claim level-transparent along $x \mapsto x^e$. The note proves (Theorems A, A′, A″, B there) that every Hodge $(2,2)$ class of every X_m^4 of *odd* degree $m \leq 199$ is algebraic.

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This paper maps the even-degree sector through degree 250 and proves its structural results. The even sector is a genuinely different landscape: the first primitive class surviving the base predicates and the complete depth-≤ 2 closure oracle occurs at $m = 50$ (so "survivor $\Rightarrow 3 \mid m$", true throughout the odd census, fails), most such classes close by *deeper* partition certificates rather than new mechanisms, and what finally remains is organized not by degree but by *claim-equivalence*. Our results:

- *The lattice is Aoki's* (Proposition 2.1): the working lattice — pair vectors and standard vectors, the $p = 2$ family included — coincides generator-by-generator with the group S_m of [3, §2], so all membership and gap verdicts are statements about Aoki's own object. The $p = 2$ standard elements are identically grade-2 Hodge quadruples, hence Lefschetz-algebraic (Lemma 2.2(i)), and the standard family is negation-closed modulo pairs (Lemma 2.2(ii)).
- *The census* (Proposition 3.1): forty primitive post-depth-two survivors through degree 250 — the tiers are defined in §3, listed one by one in Appendix A, and tabulated level by level in the ancillary — of which iterated depth-3/4 partition certificates close all but ten.
- *Exchange walls* (Theorem 4.1, Proposition 4.2): a two-entry (or three-entry) exchange with a Lefschetz-algebraic exchanger is a claim-equivalence; the exhaustive scan organizes the ten residual classes into five walls.
- *Two closures* (Propositions 5.1 and 5.4), both by the coset transfer of Proposition 2.3: the $m = 168$ class is congruent modulo S_{168} to an explicit $*$-split class, and the isolated degree-210 class lies in S_{210} itself, by an explicit 40-generator ± 1 certificate displayed in full and converted to a literal Aoki chain. Both proofs are internal to the two-paper package; Aoki's degree-form theorem and the two errata we record of its printed text (Remark 5.3) are corroboration, not support.
- *The boundary* (Corollary 6.1): eight certified gap classes in three walls; closing any member of a wall closes the wall.

Status ledger: the Propositions labeled *computer-assisted* are exhaustive machine computations with pinned receipts, engines, and independent verifiers in the ancillary files; their scope (which levels, which move family) is stated in each. Everything else is proved. The obstruction theory of the closed odd classes (exact Frobenius certificates, the negative answer to da Silva's Question 1 at $m = 33$) is in [6] and not repeated here.

2. The lattice is Aoki's S_m

Write $\nu(a)_k = \#\{i : a_i = k\}$ for $k \in \mathbb{Z}/m \smallsetminus \{0\}$ (the multiplicity vector, so ν is a bijection from finite multisets of nonzero residues onto the non-negative orthant of $\mathbb{Z}^{m-1}$), $D_m \subseteq S_m$ for the span of the pair vectors $\nu(\{k, m-k\})$, and $\mathfrak{D}_m^s$ for Aoki's classes of $s/2 + 1$ vanishing pairs (Appendix C), $\mathfrak{D}_m = \bigcup_s \mathfrak{D}_m^s$. S_m denotes Aoki's lattice: the subgroup of $\mathbb{Z}^{m-1}$ generated by the $\nu(\delta)$ over the vanishing pairs $\delta = \{k, m-k\}$ (the self-pair included at even m) and the $\nu(\sigma_{p,x})$ over the standard characters, for all primes $p \mid m$; membership is decided exactly by integer linear algebra (Hermite normal form), and in lattice statements we freely identify a with $\nu(a)$ — statements about the formal calculus, not about the character group. The first result pins the object.

Proposition 2.1 (the implemented lattice is Aoki's S_m)**.** *Aoki [3, §2] defines A_m as the zero-sum sublattice of the free abelian group on $\mathbb{Z}/m \smallsetminus \{0\}$; u — our ν — as the multiplicity-vector map, additive under juxtaposition, $\nu(\alpha * \beta) = \nu(\alpha) + \nu(\beta)$; the standard elements, verbatim, as $\sigma_{p,a} = (a,\, a + \frac{m}{p}, \ldots, a + \frac{(p-1)m}{p},\, m - pa)$ for $p \geq 3$ and $\sigma_{2,a} = (a,\, a + \frac{m}{2},\, m - 2a,\, \frac{m}{2})$ for*

$p=2$, over primes $p \mid m$ ($m>p$) and $0<a<\frac{m}{p}$ with nonzero entries; $\mathfrak{S}_m$ as the set of α admitting $\alpha * \delta \sim \sigma_1 * \cdots * \sigma_k * \delta'$ with δ, δ' unions of vanishing pairs, with $\mathfrak{D}_m \subset \mathfrak{S}_m$; and $S_m \subset A_m$ as the subgroup generated by $\nu(\mathfrak{S}_m)$. Consequently S_m equals the $\mathbb{Z}$-span of the pair vectors $\nu(\{k, m-k\})$ and the standard vectors $\nu(\sigma_{p,a})$, $p=2$ included — exactly the generator list of the implemented lattice — so the machine verdicts $\nu \in S_m$ / $\nu \notin S_m$ are statements about Aoki's group itself.

Proof. $\supseteq$: every pair class lies in $\mathfrak{D}_m \subset \mathfrak{S}_m$, and every standard element σ satisfies $\sigma * \delta \sim \sigma * \delta$, so $\sigma \in \mathfrak{S}_m$: each listed generator is in $\nu(\mathfrak{S}_m)$. $\subseteq$: if $\alpha * \delta \sim \sigma_1 * \cdots * \sigma_k * \delta'$, then $\nu(\alpha) = \sum_i \nu(\sigma_i) + \nu(\delta') - \nu(\delta)$, an integer combination of the listed generators. The two spans coincide. $\square$

Lemma 2.2 ($p=2$ standards; negation closure)**.** (i) *Every $\sigma_{2,x}$ is a grade-2 Hodge quadruple identically in x, so $V(\sigma_{2,x})$ is algebraic by Lefschetz $(1,1)$ on the Fermat surface X_m^2: the $p=2$ generators carry the same unconditional algebraicity as the odd-p standards (Theorem 2-1 of [1], with the inflation lemma of [6] when $\gcd(x, m/p) > 1$).*

(ii) *For every prime $p \mid m$ and admissible x, $\sigma_{p,x} \uplus \sigma_{p,\,m/p-x}$ is a union of vanishing pairs — $p+1$ of them for odd p, four for $p=2$; in the lattice, $-\nu(\sigma_{p,x}) \equiv \nu(\sigma_{p,\,m/p-x}) \bmod D_m$, so an integer membership certificate converts to one with nonnegative standard coefficients at the cost of pair classes.*

Proof. (i) For odd t write $r = \langle tx \rangle$; $t \cdot \frac{m}{2} \equiv \frac{m}{2}$, and the residues of $t \cdot \sigma_{2,x}$ are r, $r+\frac{m}{2}$, $m-2r$, $\frac{m}{2}$ if $r<\frac{m}{2}$ and r, $r-\frac{m}{2}$, $2m-2r$, $\frac{m}{2}$ if $r>\frac{m}{2}$ ($r=\frac{m}{2}$ is excluded: $x+\frac{m}{2} \not\equiv 0$) — both sum to $2m$. Grade-2 Hodge quadruples are the $(1,1)$ characters of X_m^2; the rational part of their Galois block is divisorial by Lefschetz $(1,1)$, whence the claim. (ii) For odd p, $d = m/p$: the translates pair off as $(x+jd) + \big((d-x)+(p-1-j)d\big) = pd = m$ for $j = 0, \ldots, p-1$, and $(m-px)+(px) = m$. For $p=2$: $\{x, m-x\}$, $\{x+\frac{m}{2}, \frac{m}{2}-x\}$, $\{m-2x, 2x\}$, and the self-pair $\{\frac{m}{2}, \frac{m}{2}\}$. Each pair lies in the span of the pair vectors. $\square$

Proposition 2.3 (coset transfer)**.** *Let α be a Hodge character of X_m^4 and let β be either a Hodge character of X_m^4 or the empty multiset $\varnothing$, for which $\nu(\varnothing) := 0$ and* claim$(\varnothing)$ *is vacuously true. If $\nu(\alpha) - \nu(\beta) \in S_m$ and* claim(β) *holds, then* claim(α) *holds. (For $\beta = \varnothing$ this is Aoki's* [3, Lemma 4.1(iii)]*, $\nu(\alpha) \in S_m \Rightarrow$* claim$(\alpha)$ *— the case used in Proposition 5.4.)*

Proof. Write $\nu(\alpha) - \nu(\beta) = \sum_i c_i\, \nu(\sigma_i) + \sum_j d_j\, \nu(\delta_j)$ with $c_i, d_j \in \mathbb{Z}$, the σ_i standard characters and the δ_j vanishing pairs (Proposition 2.1). By the negation closure of Lemma 2.2(ii), $-\nu(\sigma_{p,x}) \equiv \nu(\sigma_{p,\,m/p-x})$ modulo the span D_m of the pair vectors, so every negative standard coefficient may be traded for a positive one at the cost of pair vectors; negative pair coefficients are moved to the other side. This puts the relation in the form

$$\nu(\alpha) + \nu(\delta) \;=\; \nu(\beta) + \sum_k \nu(\sigma'_k) + \nu(\delta'),$$

with all terms multiplicity vectors of actual multisets (δ, δ' unions of vanishing pairs). Since ν is injective on multisets of nonzero residues, this is the multiset identity $\alpha * \delta \sim \beta * \sigma'_1 * \cdots * \sigma'_r * \delta'$. Now claim$(\beta)$ holds by hypothesis, claim(σ'_k) by [1, Thm. 2-1] for odd p — with the inflation lemma of [6] (Appendix B) when $g := \gcd(x, d) > 1$, $d = m/p$, since $\sigma_{p,x}$ is then the g-inflation of $\sigma_{p,x/g}$ at level m/g, where the gcd is 1 and all entries are nonzero; the reduction is never degenerate, because $g \le x < d$ forces $d/g \ge 2$ and hence $m/g = p\,(d/g) > p$ — and by Lemma 2.2(i) for $p=2$, and claim(δ') by [1, Thm. 1-1]; hence claim$(\alpha * \delta)$ by iterated application of [1, Thm. 1-4(i)] — every admitted block has even length, so at each juxtaposition r, s are even with $n = r+s+2$, and grades add, so every partial juxtaposition is again a Hodge character

— and claim(α) by [1, Thm. 1-4(ii)], $\delta \in \mathfrak{D}_m$. (Any factor that is empty is simply omitted: if $\delta' = \varnothing$ there is no $\mathfrak{D}_m$-factor on the right, and if $\delta = \varnothing$ then $\alpha * \delta = \alpha$ and the final cancellation step is not performed.) □

Remark 2.4 (even-m standard admissibility). (Two distinct conditions occur in [1] and should not be conflated: the §1 *definition* of $\sigma_{p,x}$ carries the admissibility hypothesis $d/\gcd(x, d) > 2$, while Theorem 2-1 — which exhibits the representing subvariety — assumes $\gcd(x, d) = 1$; the gap between them is bridged by the inflation lemma, as in Proposition 2.3.) For even m and odd $p \mid m$ Aoki's 1987 admissibility hypothesis $d/\gcd(x, d) > 2$ ($d = m/p$) is strictly stronger than "entries nonzero": it excludes $x \equiv d/2 \pmod d$. Every excluded multiset contains a vanishing pair — $(x + k_1 d) + (x + k_2 d) \equiv 0 \pmod m$ reduces to $1 + 2j + k_1 + k_2 \equiv 0 \pmod p$ for $x = d/2 + jd$, solvable with $0 \le k_1 < k_2 \le p - 1$ — so it is decomposable, hence algebraic for the trivial reason, and the census classifier (which tests decomposability first) is unaffected. The lattice of Proposition 2.1 follows the range of [3, §2] (nonzero entries only), matching his definition of S_m.

Remark 2.5 (the gap group). Aoki's Theorem 2.1 [3] (after [2, Thm. D]; the gap-group structure goes back to Yamamoto [13]) proves $2\alpha \in S_m$ for *every* Hodge class $\alpha \in B_m$: the gap group B_m/S_m is an elementary abelian 2-group, generated by level-raised gap generators ξ_d. Every "$2u \in S_m$" certificate below therefore instantiates a general theorem; the content of a *gap* verdict is the non-membership half $u \notin S_m$.

3. The even census

For each even m: enumerate grade-3 Hodge multisets (meet-in-the-middle on half-unit sum profiles); reduce to Galois orbits of sorted representatives; classify decomposable → quasi-decomposable → standard → $*$-split (the transport theorem of [6], stated there for all m) → survivors. The census-completeness lemma of [6] is stated and proved there for both parities (the even self-pair $m/2$ included in $M_m(1)$); the even-parity engine is calibrated against direct brute force — exact match of representative counts for $m \le 14$ — while the independent blind reimplementations of [6] cover the odd census. Throughout, *primitive* means content $\gcd(a_0, \ldots, a_5, m) = 1$, i.e. not an inflation from a lower level; eigenvalue-field labels are *numerically indicated fields* (an mpmath heuristic over finitely many split primes), used as descriptors only; they are never called eigenvalue fields without the qualifier and no formal statement rests on them. Two engine-tier facts are recorded for transparency. The receipt engine (the v3 pipeline over the even-safe base library) is multiplicity-aware throughout: its decomposability requires both members of a vanishing pair, the self-paired $m/2$ requiring multiplicity two, and its quasi loop includes the self-pair $\{m/2, m/2\}$ (e.g. $(1, 3, 36, 38, 64, 68)$ at $m = 70$ is genuinely quasi-decomposable through the self-pair, and the engine closes it). The odd-heritage display engine and the first version of the independent reimplementation carried set-based tests unsound at even m — a single $m/2$ reads as a vanishing pair; regression class $(1, 29, 35, 43, 45, 57)$ at $m = 70$, neither decomposable nor quasi-decomposable. The independent reimplementation is fixed (multiplicity-aware), both regression classes are asserted afresh by the runner, and its fresh key-level runs reproduce the engine receipt's representative and base-closure counts exactly at $50, 70, 110, 114, 168$ (pinned receipt). Independently of any engine, an exhaustive self-pair-inclusive quasi test — every pair, every split, all units — certifies all ten residual-table classes non-quasi (pinned receipt): no wall is affected. For self-containedness we recall the imported transport statement: *a Hodge* $(2, 2)$ *character of* X_m^4*, m arbitrary, whose* 6*-multiset splits into two zero-sum triples has algebraic rational Hodge classes in its Galois block* ([6, Thm. A]; its transport uses only the Galois-stable block, Lefschetz $(1, 1)$, and the $\mathbb{Q}$-linear joining-line correspondence, so no rational point on the curve and no parity hypothesis enters). *Terminology* (used consistently below): an *ordered character* is a tuple $a \in (\mathbb{Z}/m)^6$; its *sorted representative* is the sorted multiset; a *class* is the Galois orbit of such a representative (equivalently its canonical least sorted conjugate); the *rational block* $M(a)$ is the sum of the eigenspaces over the orbit; a *Hodge class* is a cohomology class; and $\nu(a)$ is the *multiplicity vector*, the object all lattice statements are about. A *base survivor*, *post-depth-two survivor*, *deep survivor* and *final unresolved class* are the successive tiers defined in this paragraph.

The closure operator. T *closes at depth* k if $T \uplus (k$ vanishing pairs) partitions into blocks each of which is a vanishing pair, a grade-2 Hodge quadruple (Lefschetz $(1,1)$), a standard block $\sigma_{p,x}$ with $p \in \{2,3,5\}$, $p \mid m$, and nonzero entries ([1, Thm. 2-1] for $p = 3,5$, with the inflation lemma when $\gcd(x, m/p) > 1$; Lemma 2.2(i) for $p = 2$) — these have four entries for $p = 2,3$ and six for $p = 5$, so the admitted block lengths are exactly $2,4,6$ — or a grade-3 sextuple already certified. Then claim(T) follows by Theorems 1-4(i),(ii) of [1].

The family is bounded, and *two tiers must be distinguished*. The *depth-k oracle* admits as blocks only pairs, grade-2 Hodge quadruples, the standard blocks with $p \leq 5$ above, and grade-3 sextuples certified by the four base predicates — block lengths exactly $2,4,6$. Standard blocks with $p \geq 7$ (length $p+1 \geq 8$) enter only the *lattice* certificates of §5, never the partition oracle; the restriction changes nothing in the tier reported here: an exhaustive sweep over all forty classes and every prime $p \geq 7$ dividing their level finds no closure by such a block at depth ≤ 2, nor at depth ≤ 3 for the residual ten (`long_standard_sweep.py` and its receipt in the ancillary). The deeper negative campaigns behind the walls run to depth 4–5 and were not re-run with long standard blocks; the statement is therefore about the headline tier and the depth-≤ 3 residual check, not about those sweeps. The *deep tiers* add depth $k \leq 5$ and *iteration*: a grade-3 block may itself carry a depth-≤ 2 certificate (two-generation certificate trees). The headline tier below is the *non-iterated* depth-≤ 2 oracle; iteration enters only in the deep tiers, where the depth and the generation used are recorded per class in the ledger. *No other family member is claimed:* no standard block other than the $\sigma_{p,x}$ above, no depth beyond the stated bounds, and no third generation. The two tiers are genuinely different counts: admitting iteration already at depth ≤ 2 would close a further part of the headline tier. Every statement below is about the *non-iterated* tier, which is exactly what `even_tier_table.py` computes.

Counting convention (the tiers are nested and each printed number names exactly one of them). *Base survivors*: orbits failing all four base predicates. *Post-depth-two survivors*: base survivors that the closure operator does not close with $k \leq 2$ added vanishing pairs. The paper's headline count $40 = 21+14+5$ is the number of *primitive post-depth-two survivors* over $m \leq 108$, $108 < m \leq 200$, $200 < m \leq 250$ (Appendix A lists all forty with their routes); these are the classes handed to the deep tiers and to the exchange analysis. The base tier is much larger and begins much earlier: base survivors first occur at $m = 32$ (e.g. $(1,9,17,20,24,25)$, which the closure operator disposes of at depth 2), and there are 1349 of them over even $m \leq 250$; all but 85 of those — the forty primitive classes counted here together with 45 induced copies — close at depth ≤ 2. Both tiers, level by level, with the canonical hash of each level's survivor list, ship as `even_tier_table.json`. The standard runner recomputes the tier predicates *from the definitions* for every even $m \leq 60$ and checks the stored lists, counts and canonical hashes at every even $m \leq 250$; a complete from-scratch regeneration of all 123 levels is a separate multi-hour command (`--emit`) and is not part of the smoke runner. Every deep closure is stored as an explicit route in the deep-closure ledger and re-verified by the standalone checkers.

Proposition 3.1 (even census)**.** (Computer-assisted; engines, closure ledger, and receipts in the ancillary.) *(i) For even $m \leq 108$ there are twenty-one primitive post-depth-two survivors, the first at $m = 50$, namely $(1,7,27,30,41,44)$, with indicated eigenvalue fields up to degree 32; the odd law "survivor $\Rightarrow 3 \mid m$" fails in the even sector. (Base survivors begin already at $m = 32$; every* *one of them below $m = 50$ closes at depth ≤ 2, and of the 1349 base survivors over even $m \leq 250$ all but 85 do — those 85 being the forty primitive classes counted here and 45 induced copies.) (ii) Eighteen of the twenty-one close by depth-$3/4$ partition certificates (two closed sextuples, or quadruple–standard–quadruple); iterating the closure operator (blocks certified by their own certificates — two-generation trees) closes $m = 54$ and $m = 90$ as well, so exactly one even class of degree ≤ 108 resists: $m = 70$, $(1, 20, 24, 42, 61, 62)$, indicated eigenvalue field $\mathbb{Q}(\zeta_{35})$. (iii) The census through degree 200 yields fourteen further primitive post-depth-two survivors, of which the same machinery closes eight (every open row in 178–200 is an induced copy of an already-closed class); the residual at $m \leq 200$ is seven classes (70; two at 110; three at 114; one at 168) and their induced copies. (iv) Through degree 250 exactly five more primitive post-depth-two survivors appear: four at 210 — $(1, 43, 44, 169, 185, 188)$, $(1, 79, 109, 121, 151, 169)$, $(2, 9, 129, 142, 168, 180)$, $(2, 82, 111, 141, 142, 152)$ — and $(1, 62, 111, 142, 162, 182)$ at 220; the deep tiers close the first and the last of the four at 210, so the residual ten of §4 is unchanged.*

Among the eighteen closures of (ii) is the $m = 66$, $\mathbb{Q}(\sqrt{-3})$ class — closed through the self-paired element $(33, 33)$, the same resource that closes da Silva's $m = 33$ witness at level 66 in [6, Thm. A″] — so at that census depth every known Fermat-fourfold Hodge class with imaginary-quadratic *indicated* eigenvalue field was algebraic; the deeper census produces a third such class at $m = 168$, closed in §5. That survivor, $(1, 25, 79, 121, 127, 151)$, has indicated eigenvalue field $\mathbb{Q}(\sqrt{-3})$, and, numerically, $p = 673$ appears as a kernel prime of its Frobenius character, so the field-of-definition obstruction of [6, Thm. C] does not fire at that prime (a computational observation, not part of any statement here).

4. Exchange and the wall structure

Theorem 4.1 (Exchange)**.** *Let $T = S \uplus A$ and $T' = S \uplus B$ be Hodge sextuples* mod m *(entries nonzero) with $|A| = |B| = 2$, and suppose $q := A \uplus (-B)$ has nonzero entries and is a Hodge $(1, 1)$ quadruple. Then* $\operatorname{claim}(T') \Rightarrow \operatorname{claim}(T)$*; since $-q$ serves the reverse direction, T and T' are* claim-equivalent*. The same holds for $|A| = 3$ with q a sextuple whose claim is known.*

Proof. $V(q)$ is algebraic by Lefschetz $(1, 1)$ on the Fermat surface X^2_m (no membership in $\mathfrak{D}_m$ is required). By Theorem 1-4(i) of [1] (juxtaposition join, r, s even), $\operatorname{claim}(T')$ and $\operatorname{claim}(q)$ give $\operatorname{claim}(T' * q)$. As multisets, $T' \uplus q = T \uplus (y_1, -y_1) \uplus (y_2, -y_2) = T * \delta$ with $\delta = (y_1, -y_1, y_2, -y_2)$ decomposable, so $\delta \in \mathfrak{D}^2_m$ and Theorem 1-4(ii) of [1] cancels it. For $|A| = 3$ the same two steps apply with q a sextuple of known claim and $\delta = B \uplus (-B)$ three vanishing pairs, $\delta \in \mathfrak{D}^4_m$ ($s = 4$ even); and $\operatorname{claim}(-q) = \operatorname{claim}(q)$ since $-q = (-1) \cdot q$ is a Galois conjugate. (Landing on a Galois conjugate $t \cdot T'$ suffices, claim being orbit-level; the $(m/2, m/2)$ self-pair is a legitimate multiset entry.) □

Proposition 4.2 (wall structure through degree 250)**.** (Computer-assisted: exhaustive exchange scan, both $|A|$ shapes, own levels and level-lifts.) *The two $m = 110$ classes are claim-equivalent — the class with numerically indicated field $\mathbb{Q}(\zeta_5)$ stands or falls with the one with numerically indicated field $\mathbb{Q}(\zeta_{110})$ — and the three $m = 114$ classes form a complete triangle at Galois-orbit level, while $m = 70$ and $m = 168$ are exchange-isolated. The three primitive classes of Proposition 3.1(iv) are already placed by the exchange graph: $(2, 9, 129, 142, 168, 180)$ at 210 is equivalent to the 3-lift of the $m = 70$ class; the 220 class is equivalent to the 2-lifts of the $m = 110$ pair; $(1, 79, 109, 121, 151, 169)$ at 210 is new and isolated. The residual boundary after the census machinery — before the closures of Propositions 5.1 and 5.4 — is thus ten primitive classes organized in five claim-equivalence walls*

$$W_{70} = \{70 \simeq 210\#2\}, \qquad W_{110} = \{110\#1 \simeq 110\#2 \simeq 220\},$$
$$W_{114} = \{\#1 \simeq \#2 \simeq \#3\}, \qquad W_{168}, \qquad W_{210} = \{210\#1\},$$

and closing any member of a wall closes the wall (Theorem 4.1). One edge in full, so that a reader can check a wall by hand: the 3*-lift of the* $m = 70$ *champion is* $(3, 60, 72, 126, 183, 186)$ *at level* 210*, and the Galois conjugate* $47 \cdot (2, 9, 129, 142, 168, 180)$ *is* $(3, 60, 94, 126, 164, 183)$*; they share* $S = (3, 60, 126, 183)$*, so* $A = (72, 186)$*,* $B = (94, 164)$ *and the exchanger*

$$q = A \uplus (-B) = (46, 72, 116, 186)$$

*has nonzero entries and is a grade-*2 *Hodge quadruple, hence Lefschetz-algebraic — a claim-equivalence between the two classes by Theorem 4.1. Precisely: call* $T \leftrightarrow T'$ *an* exchange move *when* $T = S \uplus A$*,* $T' = S \uplus B$ *with* $|A| = |B| \in \{2, 3\}$ *and the exchanger* $q = A \uplus (-B)$ *has nonzero entries and is a grade-*2 *Hodge quadruple (*$|A| = 2$*; unconditional by Lefschetz) or a grade-*3 *sextuple of known claim (*$|A| = 3$*). The scan enumerated* every *move out of* every *open class — all entry positions, all* B *— at its own level and its lifts within the census range (pinned receipts: the own levels* $70, 110, 114, 168, 210$ *and the lifted levels* $140, 220, 228$*); the open set supplied to the scan at each level is the complete final list of unresolved orbits there, so an* $|A| = 3$ *exchanger outside it carries a known algebraicity claim; each lands on the class's own wall, never on a closed class. Within this move family and level range, the transitive exchange closure of each open class is therefore exactly its wall.*

5. Two closures

Proposition 5.1 (the $m = 168$ class closes)**.** *The class* $(1, 25, 79, 121, 127, 151)$ *on* X^4_{168} *is algebraic.*

Proof. The argument is internal to the two-paper package — it uses the $*$-split transport theorem of [6] and nothing from Aoki's degree-form theorem: an explicit $*$-split witness at level 168 plus the coset transfer of Proposition 2.3. Put

$$a := (1, 25, 79, 121, 127, 151), \qquad \beta := (6, 54, 60, 108, 126, 150) \qquad \text{on } X^4_{168}$$

(the 6-fold level raising of $(1, 9, 18, 10, 21, 25)$ from level 28; only its displayed properties are used). Then, all verified exactly in the ancillary:

(a) both a and β have nonzero entries and coordinate sum $504 = 3 \cdot 168$, with constant grade 3 over all 48 units, so both are Hodge $(2, 2)$ characters of X^4_{168} — the two hypotheses of Proposition 2.3;

(b) β splits into two zero-sum triples, $6 + 54 + 108 = 168$ and $60 + 126 + 150 = 336$, so claim(β) holds by the imported $*$-split transport theorem [6, Thm. A] — unconditionally, and with no parity or rational-point hypothesis;

(c) $\nu(a) - \nu(\beta) \in S_{168}$, by an exact Hermite-normal-form certificate, while $\nu(a) \notin S_{168}$ and $\nu(\beta) \notin S_{168}$ individually — so the congruence is not vacuous and a remains a genuine gap class.

Proposition 2.3 applied to $\alpha = a$, β now gives claim(a). No Fermat-internal route of this paper or of [6] reaches a directly: its multiset admits no zero-sum-triple split (machine-checked), so the $*$-split theorem does not fire on a itself; $5 \nmid 168$ leaves no grade-3 standard block for the two-pair genre, and the one longer standard family present at this level is excluded by a residue count: the first seven entries of $\sigma_{7,x} = (x, x + 24, \ldots, x + 144, 168 - 7x)$ all lie in the class $x \bmod 24$, while the entries of a occupy the two classes $1, 7 \bmod 24$ with multiplicities $3, 3$, and each added vanishing pair contributes at most one entry to a given class (the self-pair $(84, 84)$ lies in the class 12), so at depth ≤ 2 no class reaches multiplicity 7. Within the implemented closure family, and within the stated depth, generation and lift bounds, no route to a itself was found; what closes it is the transfer from the $*$-split class β inside its own coset of S_{168}. $\square$

Remark 5.2 (the imaginary-quadratic descriptor). With this class closed, every class of the two census ranges (odd $m \leq 199$, even $m \leq 250$) whose *numerically indicated* eigenvalue field is imaginary quadratic — the three $m = 33, 66, 168$ recorded by the field-label sweep — is closed. Both the labels and the completeness of that list of three rest on the mpmath heuristic of §3, not on a theorem; the statement is recorded as an observation about the descriptor, and nothing else in this paper depends on it.

Remark 5.3 (external corroboration, and two errata of the printed source). Independently of the proof above, $168 = 2^3 \cdot 3 \cdot 7$ has the degree form of Aoki's published Theorem 0.1(i) [3], whose §4 route opens with the bridge through Corollary 3.2 there, establishing the statement for the Fermat varieties X_m^n themselves at these degrees. We do not rely on it, and record two errata of the printed text, both verified against the primary. First, the paper's own later citation of this statement ([4, Thm. 1.2(ii)], attributed to a then-preprint on Catalan curves) is a citation slip — the published Catalan paper (Tokyo J. Math. **27** (2004)) concerns the Jacobians of generalized Catalan curves, not the degree form; the theorem is [3]'s own Thm. 0.1(i). Second, in the §4 list of gap generators the $m = 28$ entry $\xi_{28} = (1, 9, 25, 12, 20, 24)$ is a misprint (it is not a zero-sum character), while the text requires $\xi_m \in B^2 \cup \big(B^4 \cap (\mathfrak{A}^1 * \mathfrak{A}^1)\big)$ and algebraizes these $V(\xi_m)$ by the proof of Theorem 4.3 of Shioda [9] (cited there); a machine-verified *possible replacement* of exactly the required join form is $(1, 9, 18) * (10, 21, 25) \in B^4_{28} \cap (\mathfrak{A}^1_{28} * \mathfrak{A}^1_{28})$, with $\nu \notin S_{28}$ and $2\nu \in S_{28}$; whether it is the intended generator modulo S_{28} is not established here. Since the closure above is internal, neither erratum bears on any statement of this paper: the 6-fold raising of that sextuple enters only as the explicit $*$-split witness β, whose two displayed properties (a),(b) are verified directly.

Proposition 5.4 (the isolated degree-210 class closes). *$V(1, 79, 109, 121, 151, 169)$ on X^4_{210} is algebraic.*

Proof. The class closes by an explicit certificate: $\nu(1, 79, 109, 121, 151, 169) \in S_{210}$, realized by 40 generators with coefficients ± 1 (re-summed independently in the ancillary; stored as a data artifact). In full, with (k) denoting the pair $\{k, 210 - k\}$:

$$\begin{aligned}\nu(a) = {}&-(7)-(8)+(12)-(21)-(27)-(28)-(29)+(36)+(41)+(42)-(61)\\&+(66)+(74)-(76)-(77)-(78)+(82)+(84)-(91)+(96)-(105)\\&+\sigma_{2,4}-\sigma_{2,9}+\sigma_{2,14}+\sigma_{2,21}+\sigma_{2,27}+\sigma_{2,28}+\sigma_{2,29}-\sigma_{2,31}-\sigma_{2,41}-\sigma_{2,42}\\&-\sigma_{3,4}+\sigma_{3,6}+\sigma_{3,7}+\sigma_{3,8}+\sigma_{3,9}-\sigma_{3,12}-\sigma_{3,14}+\sigma_{7,1}-\sigma_{7,6}\,.\end{aligned}$$

Every generator's eigenspace is algebraic — pairs: linear cycles, Theorem 1-1 of [1]; $\sigma_{3,x}, \sigma_{7,x}$: Theorem 2-1 of [1], with the inflation lemma of [6] when $\gcd(x, m/p) > 1$; $\sigma_{2,x}$: Lemma 2.2(i) — so converting the negative terms by Lemma 2.2(ii) produces the literal membership $a * \delta \sim \sigma_1 * \cdots * \sigma_{19} * \delta'$ (δ, δ' unions of vanishing pairs), and $V(a)$ is algebraic by Theorems 1-4(i),(ii) of [1] — the calculus of Lemma 4.1 of [3] (which carries no hypothesis on m), with every block certified above. □

Negative-coefficient certificates of this kind are invisible to monoid/partition search engines — the lattice membership test ($\nu(a) \in S_m$, Hermite normal form) decides standard-calculus reachability exactly and is the instrument we now run first. (The two directions, with the scope stated precisely: a certificate all of whose blocks are vanishing pairs, standard blocks, or classes already certified inside S_m forces $\nu(a) \in S_m$, since ν is additive under $*$; a certificate routed through a Lefschetz quadruple whose own ν lies outside S_m escapes the lattice — the $\mathbb{Q}(\zeta_{21})$ class at 105 closes exactly this way in [6, Thm. A′] while remaining a gap class — so gap-ness bounds the *standard* calculus, not partition search as such. Conversely an integer membership combination converts to a literal Aoki chain by negation closure (Lemma 2.2(ii)), as carried out in Proposition 5.4.)

6. The boundary at $m \leq 250$

Corollary 6.1 (the even-degree boundary at $m \leq 250$)**.** *The residual boundary for all* even *degrees $m \leq 250$ — unresolved by the implemented closure family and the cited results of the two-paper package — is eight primitive classes in three walls (W_{70}: two members, W_{110}: three, W_{114}: three, including the displayed lift-equivalent members at* 210 *and* 220*), and all eight are certified gap classes: $\nu \notin S_m$, $2\nu \in S_m$ (the doubling holds for* every *Hodge class by* [3, Thm. 2.1]*; the certificates make it explicit). The residual question is uniform: algebraicity of gap-group classes at degrees* $70, 110, 114$ *— outside the degree forms of Aoki's theorem (*70 *and* 210 *contain both* 5 *and* 7*;* 110 *contains* 11*;* 114 *contains* 19*).*

m	class	$\nu \in S_m$?	status
70	$(1, 20, 24, 42, 61, 62)$	no	unresolved (W_{70})
110	$(1, 24, 62, 71, 81, 91)$	no	unresolved (W_{110})
110	$(1, 31, 55, 71, 81, 91)$	no	unresolved (W_{110})
114	$(1, 13, 43, 72, 103, 110)$	no	unresolved (W_{114})
114	$(1, 13, 43, 80, 102, 103)$	no	unresolved (W_{114})
114	$(1, 7, 78, 79, 86, 91)$	no	unresolved (W_{114})
168	$(1, 25, 79, 121, 127, 151)$	no	closed (Prop. 5.1)
210	$(1, 79, 109, 121, 151, 169)$	yes	closed (Prop. 5.4)
210	$(2, 9, 129, 142, 168, 180)$	no	unresolved (W_{70})
220	$(1, 62, 111, 142, 162, 182)$	no	unresolved (W_{110})

The ten primitive residual classes of degree ≤ 250 after the depth-two and the deep closure stages. Every lattice verdict carries an independent certificate in the ancillary: a modular kernel witness ($q = 2$) for each "no", the re-summed ± 1 combination for the "yes".

A systematic treatment of the even sector beyond degree 250 is deferred.

7. Methods, code availability, and provenance

Exact objects throughout: multiset identities, Hodge grades, and lattice membership are decided in exact integer arithmetic; no floating point enters any certificate. Every lattice verdict has an independent exact certificate (modular kernel witnesses found by the standalone script's own elimination for each non-membership; exact re-summation for each membership and for the doubled vectors). The coset congruence of Proposition 5.1 carries both kinds: a stored integer combination over the generator list (94 coefficients in $[-3, 3]$), re-summed from disk by the standalone checker, and a modular kernel witness which — because the difference lies in S_{168} — separates $\nu(a)$ and $\nu(\beta)$ from the lattice simultaneously; the full census and exchange sweeps are reproducible from the supplied engines and are accompanied by checksummed campaign receipts; independent brute-force calibration is performed at small levels ($m \leq 14$), and the algorithmically independent reimplementation re-derives the representative counts fresh at the key levels $50, 70, 110, 114, 168$ (pinned receipt). The exact-arithmetic certificate framework is the one developed for the Fermat surfaces of [7], and the odd-degree half of the census is certified in [6] by three independent implementations, including a direct brute-force enumeration at every level $m \leq 143$. `run_all.sh` *fresh-runs* the smoke and certificate tiers, including the tier and final-table verifiers of this paper; the multi-hour exhaustive even and exchange campaigns are supplied as SHA-pinned *historical receipts* with their own regeneration commands, and are not re-executed by that runner. The author takes full responsibility for all proofs, code, and bibliographic claims. *Data and code availability.* The package is submitted as ancillary files and licensed for reuse (see `LICENSE` there). The same package is deposited as an immutable archival release under the concept DOI `10.5281/zenodo.21670577`, which always resolves to the latest version; that record carries both manuscripts of the two-paper package. The ancillary package is shared with the companion note: engines (census, closure oracle, exchange scan, lift sweeps, lattice membership), the deep-closure ledger, the stored degree-210 certificate, the complete even census receipt over all even $6 \le m \le 250$, and the pinned exchange/negative-sweep logs, with a checksum manifest and a fresh-run runner (`run_all.sh`; "PASSED" always means executed now, never cached). This manuscript was prepared with AI assistance; all computational claims are backed by the ancillary code and receipts.

Appendix A. The forty primitive post-depth-two survivors

Every primitive class over even $6 \le m \le 250$ that survives the four base predicates *and* the depth-≤ 2 closure oracle, with the tier that disposes of it: *depth-3*/*depth-4* = a deeper partition certificate, *iterated* = a two-generation certificate tree, *Prop.* = one of the two closures of §5, *open* = unresolved, with its wall. All forty have content 1; every open class satisfies $\nu \notin S_m$ and $2\nu \in S_m$. The machine-readable table, with the exact lattice verdicts and the ledger route strings, is `even_final_table.json` (regenerated and re-verified by `even_final_table.py --check`); the per-level tier counts and canonical hashes are `even_tier_table.json`.

m	class	closed by	m	class	closed by
50	$(1,7,27,30,41,44)$	depth-4	108	$(1,44,55,58,82,84)$	depth-3
54	$(1,7,19,36,49,50)$	iterated	110	$(1,24,62,71,81,91)$	open, W_{110}
54	$(1,7,24,38,41,51)$	depth-3	110	$(1,31,55,71,81,91)$	open, W_{110}
54	$(1,7,36,38,39,41)$	depth-3	114	$(1,7,78,79,86,91)$	open, W_{114}
66	$(1,8,23,52,54,60)$	depth-3	114	$(1,13,43,72,103,110)$	open, W_{114}
66	$(1,25,31,37,49,55)$	depth-3	114	$(1,13,43,80,102,103)$	open, W_{114}
70	$(1,6,43,45,57,58)$	depth-3	114	$(1,44,60,77,78,82)$	depth-3
70	$(1,20,24,42,61,62)$	open, W_{70}	138	$(1,22,57,63,135,136)$	depth-3
72	$(1,10,37,50,54,64)$	depth-3	138	$(1,70,75,81,93,94)$	depth-3
72	$(1,26,37,40,54,58)$	depth-3	140	$(1,58,71,86,90,114)$	depth-3
78	$(1,12,43,55,60,63)$	depth-3	150	$(1,39,66,101,120,123)$	iterated
78	$(1,20,43,45,62,63)$	depth-3	156	$(1,79,84,90,106,108)$	depth-3
78	$(1,28,42,45,54,64)$	depth-3	156	$(2,45,84,106,108,123)$	depth-3
90	$(1,8,26,62,85,88)$	depth-3	168	$(1,25,79,121,127,151)$	Prop. 5.1
90	$(1,23,38,55,74,79)$	iterated	168	$(1,27,87,113,126,150)$	depth-3
90	$(1,38,46,55,56,74)$	depth-3	210	$(1,43,44,169,185,188)$	depth-3
102	$(1,14,16,82,93,100)$	depth-3	210	$(1,79,109,121,151,169)$	Prop. 5.4
102	$(1,16,28,62,99,100)$	depth-3	210	$(2,9,129,142,168,180)$	open, W_{70}
102	$(1,40,52,69,70,74)$	depth-3	210	$(2,82,111,141,142,152)$	depth-3
102	$(1,46,52,57,70,80)$	depth-3	220	$(1,62,111,142,162,182)$	open, W_{110}

Appendix B. Statements imported from the companion note

For self-containedness we restate, in the numbering of [6], the four results of that note used here. Their proofs are there; nothing in this paper re-proves them, and every use is cited at the point of use.

– **Theorem A ($*$-split transport).** *Let a be a Hodge $(2,2)$ character of X^4_m, m arbitrary, whose 6-multiset splits as two zero-sum triples $\beta' \uplus \gamma'$. Then the rational Hodge classes of $M(a)$ are algebraic: each is the image, under the algebraic joining-line correspondence of the Shioda–Katsura structure map, of a $\mathbb{Q}$-linear combination of divisor classes on $X^1_m \times X^1_m$ furnished by Lefschetz $(1,1)$.* (Used for β in Proposition 5.1 and as the fourth base predicate.)

– **Lemma (inflation).** *Let $m = e\,m'$ and β a level-m' character with every entry of $e\beta$ nonzero mod m. Then* $\mathrm{claim}_{m'}(\beta) \Rightarrow \mathrm{claim}_m(e\beta)$. **Lemma (descent).** *Under the same hypotheses,* $\mathrm{claim}_m(e\beta) \Rightarrow \mathrm{claim}_{m'}(\beta)$. Both are proved there from $\pi\colon X^n_m \to X^n_{m'}$, $(x_i) \mapsto (x_i^e)$, finite and surjective, via $\pi_*\pi^* = (\deg\pi)\,\mathrm{id}$; the nonzero-entry hypothesis is what makes $e\beta$ a legitimate character. Note the hypothesis is on the *entries*, not on $\gcd(x, m/p)$: a standard block $\sigma_{p,x}$ with $g = \gcd(x,d) > 1$, $d = m/p$, is the g-inflation of $\sigma_{p,x/g}$ at level m/g, where the gcd is 1, and all its entries are nonzero, so the inflation lemma applies verbatim.
– **Lemma (census completeness).** *For each m and the shipped generator, classifiers and canonizer: every element of M_m has even length $2g$ with g its grade; $M_m(1)$ consists exactly of the vanishing pairs $\{k, m-k\}$, the self-pair $k = m/2$ included at even m; a grade-3 sextuple is decomposable exactly when it contains a vanishing pair; for a pair-free sextuple, quasi-decomposability is equivalent to some $a \uplus \{k, m-k\}$ splitting into two grade-2 Hodge quadruples; the meet-in-the-middle generator produces exactly the grade-3 Hodge sextuples; and canonization by the least sorted conjugate yields one representative per Galois orbit.* Stated and proved there for *both* parities.

Appendix C. The cited Aoki statements

For self-containedness we reproduce the exact statements used, in the numbering of the primary texts.

From [1] (notation as there: $\mathfrak{B}^n_m$ the Hodge characters of X^n_m, $\mathfrak{D}^n_m$ the classes that are coordinate permutations of $(a_0, -a_0, \ldots, a_r, -a_r)$, $r = n/2$; $*$ juxtaposition; $\textsc{claim}(\alpha)$: $V(\alpha)$ is generated by classes of algebraic cycles on X^n_m):

– **Theorem 1-1** (Shioda). *The linear space L represents $\delta \in \mathfrak{D}^n_m$; more precisely $\omega_\delta(L) \cdot \overline{\omega_\delta(L)} = (-1)^r m^{n+1}$.*
– **Theorem 1-4** (Shioda [8], Ran [10]). *Let r and s be non-negative even integers such that $n = r + s + 2$, and let $\alpha \in \mathfrak{B}^r_m$, $\beta \in \mathfrak{B}^s_m$. Then: (i) if $\textsc{claim}(\alpha)$ and $\textsc{claim}(\beta)$ are true, then $\textsc{claim}(\alpha * \beta)$ is also true; (ii) if there exists $\delta \in \mathfrak{D}^s_m$ such that $\textsc{claim}(\alpha * \delta)$ is true, then $\textsc{claim}(\alpha)$ is also true.*
– **Theorem 2-1.** *For an odd prime $p \mid m$, $d = m/p$, $\gcd(a,d) = 1$, the variety Y defined by (2.1) there is a subvariety of X^{p-1}_m of codimension $r = (p-1)/2$ representing $\alpha = \sigma_{p,a}$; more precisely $\omega_\alpha(Y) \cdot \overline{\omega_\alpha(Y)} = (-1)^r p^{p-2} m^p$.*

From [3] (verified verbatim against the open-access copy in the Rikkyo University repository, doi:10.14992/00009819; §5 records two errata of the printed text, both likewise verified against the primary):

– **§2 definitions** (verbatim). *If m is divisible by a prime number p and $m > p$, then we have explicitly defined elements of $\mathfrak{B}_m$: $\sigma_{p,a} = (a,\, a + \frac{m}{p},\, \ldots,\, a + \frac{(p-1)m}{p},\, m - pa)$ if $p \geq 3$; $(a,\, a + \frac{m}{2},\, m - 2a,\, \frac{m}{2})$ if $p = 2$ — where a is an integer such that $0 < a < \frac{m}{p}$. We call $\sigma_{p,a}$ a standard element.* Further (§2 there): A_m is the zero-sum sublattice of the free abelian group on $\mathbb{Z}/m \setminus \{0\}$; u (our ν) the multiplicity-vector map, $\nu(\alpha * \beta) = \nu(\alpha) + \nu(\beta)$; $B_m, S_m, D_m \subset A_m$ the subgroups generated by $u(\mathfrak{B}_m), u(\mathfrak{S}_m), u(\mathfrak{D}_m)$, with $\mathfrak{D}_m \subset \mathfrak{S}_m \subset \mathfrak{B}_m$ and $\mathfrak{S}_m$ the set of α admitting $\alpha * \delta \sim \sigma_1 * \cdots * \sigma_k * \delta'$ $(\delta, \delta' \in \mathfrak{D}_m)$.
– **Theorem 0.1.** *Suppose the prime factorization of m is one of the following forms: (i) $m = 2^a 3^b 5^c 7^d$, where a, b, c, d are non-negative integers such that either $c = 0$ or $d = 0$; (ii) $m = p^e$ or $2p^e$, p an odd prime. Then the Hodge conjecture is true for all abelian varieties of Fermat type of degree m.* Its proof (§4 there, p. 185) opens: "In view of Corollary 3.2, it suffices to show that the Hodge conjecture for X^n_m is true for

all n" — the bridge to the Fermat varieties themselves used in Proposition 5.1; the verification then reduces, via Lemma 4.1 and Theorem 2.1 there, to the gap generators ξ_m at $m = 12, 15, 20, 21, 28$ (the $m = 28$ entry being the misprint recorded in §5).

- **Theorem 2.1** (verbatim; proved as Theorem D of [2]). *For each divisor $d > 1$ of m such that $\mu'(d) = 1$, there exists an element ξ_d of $B_d - S_d$ with the property that the Γ_m-module B_m is generated by $(m/d)\xi_d$ for all such divisors d of m and S_m. Moreover, we have $2\alpha \in S_m$ for every $\alpha \in B_m$.* (μ' is Aoki's modified Möbius function and (m/d) his level-raising product map; their exact forms are not needed here.)
- **Lemma 4.1** (verbatim). *Let $\alpha, \beta \in B_m$. Then: (i) if both $V(\alpha)$ and $V(\beta)$ are algebraic, then so is $V(\alpha * \beta)$; (ii) if $V(\alpha * \delta)$ is algebraic for some $\delta \in D_m$, then so is $V(\alpha)$; (iii) if $\alpha \in S_m$, then $V(\alpha)$ is algebraic.* No hypothesis on m is imposed; Proposition 2.1 identifies the implemented lattice with S_m generator-by-generator.

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Email address: `jum.rifm@gmail.com`